Quiz2_solutions

Logistic Regression with the Titanic Data – Classification Task

we exclude PassengerID, Name, Ticket, and Cabin in the ensuing analysis. Next, please note that Age has many missing values – my suggestion is to delete those with missing values.

```
if (!requireNamespace("tidyverse")) install.packages('tidyverse')
```

```
## Loading required namespace: tidyverse
```

```
if (!requireNamespace("caret")) install.packages('caret')
```

```
## Loading required namespace: caret
```

```
if (!requireNamespace("leaps")) install.packages('leaps')
```

```
## Loading required namespace: leaps
```

```
if (!requireNamespace("bestglm")) install.packages('bestglm')
```

```
## Loading required namespace: bestglm
```

```
if (!requireNamespace("MASS")) install.packages('MASS')
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3

## Warning: package 'tibble' was built under R version 4.4.1

## Warning: package 'tidyr' was built under R version 4.4.1

## Warning: package 'purrr' was built under R version 4.4.1

## Warning: package 'lubridate' was built under R version 4.4.1

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.1     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.4.1

## Loading required package: lattice

## Warning: package 'lattice' was built under R version 4.4.1

##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```r
library(leaps)
library(bestglm)
library(MASS)
```

```
## Warning: package 'MASS' was built under R version 4.4.1

##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select
```

```r
# 1)
data = read.csv('Titanic2.csv')
data = data[,-c(1,4,9,11)]
data = na.omit(data)
cat('There are', nrow(data), 'observations left.')
```

```
## There are 714 observations left.
```

```r
# 2)
str(data)
```

```
## 'data.frame':    714 obs. of  8 variables:
##  $ Survived: int  0 1 1 1 0 0 0 1 1 1 ...
##  $ Pclass  : int  3 1 3 1 3 1 3 3 2 3 ...
##  $ Sex     : chr  "male" "female" "female" "female" ...
##  $ Age     : num  22 38 26 35 35 54 2 27 14 4 ...
##  $ SibSp   : int  1 1 0 1 0 0 3 0 1 1 ...
##  $ Parch   : int  0 0 0 0 0 0 1 2 0 1 ...
##  $ Fare    : num  7.25 71.28 7.92 53.1 8.05 ...
##  $ Embarked: chr  "S" "C" "S" "S" ...
##  - attr(*, "na.action")= 'omit' Named int [1:177] 6 18 20 27 29 30 32 33 37 43 ...
##   ..- attr(*, "names")= chr [1:177] "6" "18" "20" "27" ...
```

```r
data$Survived = as.factor(data$Survived)
data$Pclass = as.factor(data$Pclass)
data$Sex = as.factor(data$Sex)
data$Embarked = as.factor(data$Embarked)
str(data)
```

```
## 'data.frame':    714 obs. of  8 variables:
##  $ Survived: Factor w/ 2 levels "0","1": 1 2 2 2 1 1 1 2 2 2 ...
##  $ Pclass  : Factor w/ 3 levels "1","2","3": 3 1 3 1 3 1 3 3 2 3 ...
##  $ Sex     : Factor w/ 2 levels "female","male": 2 1 1 1 2 2 2 1 1 1 ...
##  $ Age     : num  22 38 26 35 35 54 2 27 14 4 ...
##  $ SibSp   : int  1 1 0 1 0 0 3 0 1 1 ...
##  $ Parch   : int  0 0 0 0 0 0 1 2 0 1 ...
##  $ Fare    : num  7.25 71.28 7.92 53.1 8.05 ...
##  $ Embarked: Factor w/ 3 levels "C","Q","S": 3 1 3 3 3 3 3 3 1 3 ...
##  - attr(*, "na.action")= 'omit' Named int [1:177] 6 18 20 27 29 30 32 33 37 43 ...
##   ..- attr(*, "names")= chr [1:177] "6" "18" "20" "27" ...
```

```r
# 3)
set.seed(2026)
training.samples = data$Survived %>%
  createDataPartition(p = 0.8, list = FALSE)
train.data  = data[training.samples, ]
test.data = data[-training.samples, ]

# 4)
model <- glm(Survived ~ . , data = train.data, family = binomial)
summary(model)
```

```
##
## Call:
## glm(formula = Survived ~ ., family = binomial, data = train.data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  4.571032   0.626131   7.300 2.87e-13 ***
## Pclass2     -1.311994   0.379553  -3.457 0.000547 ***
## Pclass3     -2.580547   0.397686  -6.489 8.65e-11 ***
## Sexmale     -2.625008   0.255539 -10.272  < 2e-16 ***
## Age         -0.043594   0.009403  -4.636 3.55e-06 ***
## SibSp       -0.403452   0.143820  -2.805 0.005028 **
## Parch       -0.097200   0.151577  -0.641 0.521353
## Fare         0.003389   0.003457   0.980 0.326983
## EmbarkedQ   -1.028851   0.626220  -1.643 0.100392
## EmbarkedS   -0.376063   0.309419  -1.215 0.224220
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 772.45  on 571  degrees of freedom
## Residual deviance: 499.64  on 562  degrees of freedom
## AIC: 519.64
```

```
##
## Number of Fisher Scoring iterations: 5
```

```r
# 5)
probabilities <- model %>% predict(test.data, type = "response")
predicted.classes <- as.factor(ifelse(probabilities > 0.5, 1, 0))
confusionMatrix(predicted.classes, test.data$Survived, positive = '1')
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0  1
##          0 68 12
##          1 16 46
##
##                Accuracy : 0.8028
##                  95% CI : (0.7278, 0.8648)
##     No Information Rate : 0.5915
##     P-Value [Acc > NIR] : 6.975e-08
##
##                   Kappa : 0.5963
##
##  Mcnemar's Test P-Value : 0.5708
##
##             Sensitivity : 0.7931
##             Specificity : 0.8095
##          Pos Pred Value : 0.7419
##          Neg Pred Value : 0.8500
##              Prevalence : 0.4085
##          Detection Rate : 0.3239
##    Detection Prevalence : 0.4366
##       Balanced Accuracy : 0.8013
##
##        'Positive' Class : 1
##
```

```r
# 6)
new = test.data[1:3,]
new <- data.frame(
  Pclass   = factor(c(3,1,2), levels = levels(train.data$Pclass)),
  Sex      = factor(c("male","female","male"), levels = levels(train.data$Sex)),
  Age      = c(24,68,41),
  SibSp    = c(1,0,1),
  Parch    = c(0,0,2),
  Fare     = c(8.42,24.34,41.93),
  Embarked = factor(c("Q","C","S"), levels = levels(train.data$Embarked))
)
(p <- model %>% predict(new, type = "response"))
```

```
##          1          2          3
## 0.04374957 0.84411266 0.12077335
```

```r
ifelse(p > 0.5, "Survived", "Not survived")
```

```
##              1              2              3
## "Not survived"     "Survived" "Not survived"
```

```r
# 7)
# Let p = P(Survived = 1 | X).
# log(p / (1 - p)) =
# 4.5710 - 1.312*I(Pclass=2) - 2.581*I(Pclass=3)
# - 2.625*I(Sex="male") - 0.0436*Age - 0.4035*SibSp
# - 0.0972*Parch + 0.00339*Fare
# - 1.0289*I(Embarked="Q") - 0.3761*I(Embarked="S")

# Pclass2, Pclass3, Sexmale, Age and SibSp are significant.

# 8)
# Passenger #893, p = 0.84411266 the largest.
```